

BOLLAPALLY BANSI DEEPAK

AI Engineer / Full Stack Developer

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Hyderabad, India

BansiDeepak

Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer and AI Engineer specializing in building end-to-end web applications and integrating AI/ML and **Computer Vision** systems. Proficient in **React**, **Next.js**, **Node.js**, and **TypeScript** for front-end and back-end development, with deep expertise in **LLM frameworks**, **LangChain**, **RAG Architectures**, and **vector search**. Experienced in deploying production-grade AI platforms using **FastAPI**, **Celery**, and **microservices**, with asynchronous processing and real-time inference at scale.

PROFESSIONAL EXPERIENCE

AI Engineer

Neurom Innovations

Dec 2024 – Present

Hyderabad, India

- Architected end-to-end RAG systems using schema-aware retrieval, enabling high-accuracy LLM responses for enterprise data sources.
- Built a dynamic multi-model inference gateway in **FastAPI** that intelligently routes traffic between **Google Gemini** and **OpenAI** based on task complexity, latency budget, and cost-efficiency.
- Built production **computer-vision** pipelines – real-time face detection, deep face-embedding models, and **pgvector** similarity search – powering a multi-tenant biometric attendance platform with **AES-256-GCM**-encrypted PII.
- Engineered real-time **geospatial** and **signal-processing** systems – live **WebGL** satellite-orbit tracking and **SDR** pipelines (FFT, FM demodulation) – on **event-driven** Node.js/TypeScript microservices.
- Reduced hallucinations by designing enterprise-grade guardrails, including policy-driven validation, tool-calling restrictions, and structured output enforcement.
- Led full-stack development of internal AI pilots, including React/Next.js dashboards, asynchronous inference queues, and real-time task monitoring.
- Delivered full-stack web applications across multiple client projects – building React/Next.js frontends, FastAPI/Express backends, and integrating AI systems end-to-end.
- Optimized LLM performance pipelines, benchmarking inference cost, throughput, caching layers, and implementing structured prompting frameworks.

KEY PROJECTS

CORAX – AI Face-Recognition Attendance Platform (In Progress)

FastAPI, PostgreSQL/pgvector, Celery, InsightFace, MediaPipe, React, React Native, Docker

- Architecting a multi-tenant face-recognition attendance platform end-to-end – an async **FastAPI** back-end (**81 REST endpoints**, 15 data models), a **React + TypeScript** dashboard, and a **React Native** mobile app – containerized as a **7-service Docker** stack (Postgres/pgvector, Redis, Celery, nginx).
- Engineered a real-time face-recognition pipeline pairing client-side **MediaPipe** 468-landmark detection (7 Hz) with a server-side two-stage stack – an **InsightFace SCRFD** detector plus a **512-dim deep face-embedding model** – and **pgvector** cosine similarity for sub-second identity matching.
- Built an asynchronous video-processing pipeline on **Celery + Redis** that returns **HTTP 202** instantly and runs frame sampling and inference off the request path, keeping the API responsive under heavy media load.
- Implemented production-grade security – **AES-256-GCM** encryption of PII/biometric data, **JWT** auth with refresh-token rotation and revocation, rate limiting, account lockout, and JWT-enforced tenant isolation.
- Developed a data-rich **React + TypeScript** dashboard with **React Query** (90+ data hooks), **Zustand**, and **Recharts** – featuring drill-down navigation, live attendance status, and CSV/PDF reporting.
- Delivered a cross-platform **React Native (Expo)** app for on-site capture – native-camera face enrollment, offline-aware sync, biometric login, and push notifications.

ASKENCE – Enterprise Text-to-SQL & Analytics Platform

React 19, TypeScript, FastAPI, LangChain, Gemini, PostgreSQL, Docker

- Designed an advanced Text-to-SQL RAG system using **LangChain** and **Schema Injection** to generate accurate and executable SQL queries from natural language.
- Built a **SQL Validation & Rewriting Engine** that detects large datasets and automatically produces optimized, aggregated queries.
- Developed a dynamic visualization engine using **React + TypeScript + Vite** that classifies output types and renders appropriate visualizations.

TECHNICAL SKILLS

React.js

Next.js

React Native

TypeScript

JavaScript

Vite

Tailwind CSS

Cesium

WebGL

Node.js

Express

FastAPI

Python

Celery

Redis

Streamlit

Google Gemini

OpenAI API

LangChain

RAG Pipelines

Computer Vision

Vector Search

Prompt Engineering

Text-to-SQL

Pandas

NumPy

Plotly

SDR/DSP

PostgreSQL

pgvector

Docker

Kubernetes

AWS Amplify

AWS EC2

Git

CI/CD

CERTIFICATIONS

1. Oracle Cloud Infrastructure 2023 Certified Associate

2. Postman API Fundamentals Student Expert

EDUCATION

B.Tech, Computer Science & Engineering

Talla Padmavathi College of Engineering

📅 2021 – 2025

- Deployed system for **real-world enterprise** data analytics, enabling business users to query operational data through natural language.

ISVT GROUND STATION – Real-Time Satellite Tracking & SDR Platform *(In Progress)*

React, TypeScript, Node/Express, PostgreSQL, Redis, Cesium, satellite.js, Python, RTL-SDR

- Building a live LEO-satellite visualizer that renders real-time orbits via client-side **SGP4** propagation (**satellite.js**) in a **Cesium/WebGL** 3D scene, paired with a 2D **Map-box** ground track and sub-second position updates.

• Engineered an **event-driven** Node/Express + **TypeScript** backend with a central **EventBus** and pluggable **connectors** that ingest TLE feeds from **Celestrak** and **Space-Track** into **PostgreSQL** with **Redis** caching and a 2-hour refresh cron.

• Developed a **Python SDR** signal pipeline (**RTL-SDR**) performing **FM demodulation**, **FFT** spectrum analysis, and live **waterfall** rendering streamed to the React frontend in real time.

• Architected a **TypeScript monorepo** (npm workspaces) with a shared type ontology, raw-SQL migrations, and a **versioned REST API** (strict-mode, consistent response envelope); implemented orbital math — pass prediction, **Doppler** shift, and topocentric elevation/azimuth across ECI/ECEF/geodetic frames.

SEARCHLIFT360 – Asynchronous AI SEO Auditing Engine

FastAPI, Python, Docker, Next.js 14, AWS Amplify

- Architected a scalable asynchronous crawling engine using **FastAPI Background Tasks**, enabling parallel deep-site audits with zero UI blocking.

• Developed a unique hybrid analysis pipeline combining **rule-based DOM Checks** with **generative AI insights** for content strategy.

• Automated the creation of client-ready PDF audit reports using **ReportLab**, generating structured, visual SEO health summaries.